higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## APPLIED ENGINEERING TECHNOLOGY
## NQF LEVEL 4

## 18 NOVEMBER 2019

**This marking guideline consists of 6 pages.**

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## QUESTION 1

1.1
1.1.1 D
1.1.2 C
1.1.3 D
1.1.4 C
1.1.5 B
1.1.6 A
1.1.7 D
1.1.8 D
1.1.9 B
1.1.10 A

(10 × 1) (10)

1.2
- Reciprocating pump
- Centrifugal pump
- Rotary pump
- Steam pump
- Piston less pump
- Liquid ring pump

(Any 5 × 1) (5)

1.3 $W = mg$ $m = 5 \times 1\,000 = 5\,000$ kg✓

$= 5\,000 \times 9{,}81$

$= 49\,050$ N✓

$W = NR$

$\mu = \frac{F_{FF}}{NR}$ ✓

$= \frac{10\,000}{49\,050}$✓

$= 0{,}204$✓ (5)

**[20]**

## QUESTION 2

2.1 REACTIONS

MOMENTS ABOUT A

ACWM = CWM
B × 6 = (30 × 2) + (10 × 4)✓
B = 16,66 kN ✓

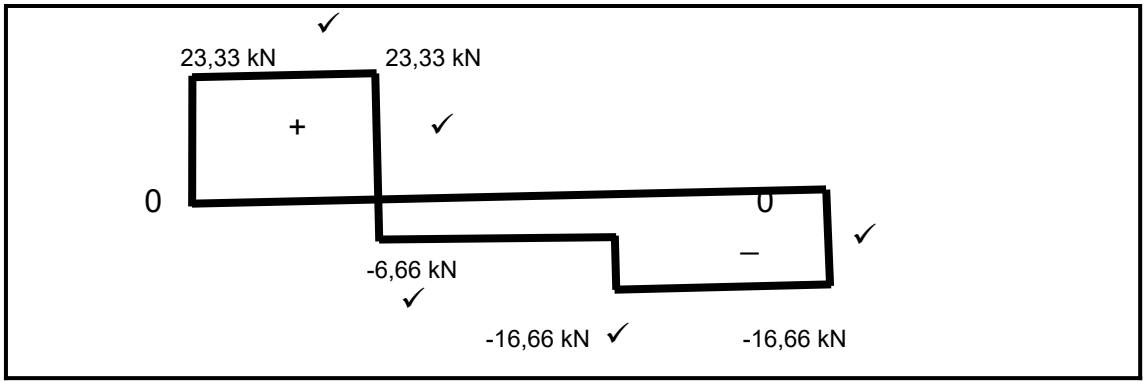
MOMENT ABOUT B

ACWM = CWM
(20 × 6) + (30 × 4) + (10 × 2) = A × 6✓
A = 43,3 kN✓

TEST

Forces ↓ = Forces ↑
20 × 30 +10 = 16,66 + 43,3
60 = 60✓ (5)

2.2




SHEAR FORCE DIAGRAM (5)

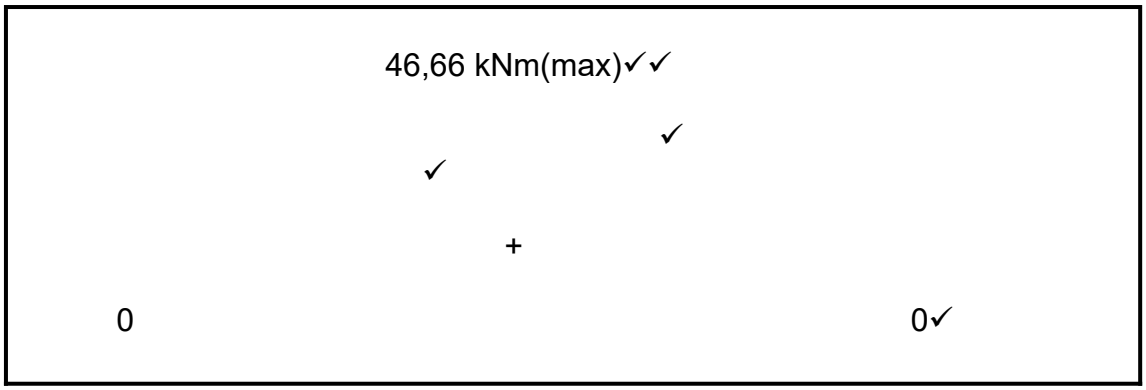
2.3 BMA = 0 kNm✓

BMB = (43,3 × 2) – (20 × 2) = 46,6 kNm✓

BMC = (16,66 × 2) = 33,3 kNm✓

BMD = 0 kNm✓ (4)

2.4




BENDING MOMENT DIAGRAM (5)

2.5 BMB = (43,3 × 2) – (20 × 2)✓✓
= 86,6 – 40✓
= 46,6 kNm (maximum bending moment)✓ (4)
**[23]**

**QUESTION 3**

3.1
3.1.1 B
3.1.2 A
3.1.3 J
3.1.4 C
3.1.5 I
3.1.6 D
3.1.7 L
3.1.8 E
3.1.9 F
3.1.10 G

(10 × 1) (10)

3.2
- An armature or motor
- A commutator
- Brushes
- An axle
- An electromagnet or solenoid (5)

**[15]**

## QUESTION 4

4.1 4.1.1 Inlet or induction stroke – As the piston moves down a vacuum is created in the combustion chamber cylinder which allows the mixture of petrol vapour and air in through the open inlet valve.

4.1.2 Compression stroke – The piston is moving upwards and with both valves closed, the mixture is compressed.

4.1.3 Power stroke – As the piston reaches close to top dead centre, the spark plug ignites the mixture. The pressure in the cylinder increases and forces the piston downwards.

4.1.4 Exhaust stroke – As the piston moves upwards the exhaust valve opens and the burned gases is forced through exhaust valve and thus clears the chamber.

(4 × 2) (8)

4.2 Force = P x A ✓

= 5 000 Pa ✓ x 0,01767 $m^2$ ✓
= 88,357 N ✓

(4)

4.3
- Make sure that the drill press has a working emergency button.
- Make sure the machine guards are not damaged.
- Make sure the machine guard is in place and functioning properly.
- Use a vacuum, brush or rake to remove cuttings.
- Remove burrs and chips from a drilled hole.
- Switch the machine off when you are finished.
- Always wear the right protective equipment. (Any 5 × 1) (5)

4.4
- Case
- Stator
- Armature
- Commutator
- Brushes (Any 3 × 1) (3)

**[20]**

## QUESTION 5

5.1
- Design specifications
- Produce concept drawings
- Evaluate the concepts
- Strength of materials
- Reliability
- Serviceable
- Manufacturability
- Disposability
- Calculations
- Build a prototype

(Any 5 × 1) (5)

5.2 5.2.1 A photoelectric smoke detector uses a light source and a sensor to detect and set off an alarm when smoke is present.

5.2.2 A speaker is used to make continuous noise, notes or speech.

5.2.3 A motor is used to create movement in either rotational direction depending on which way it is connected. It is used as an actuator.

5.2.4 Buzzers can give a continuous sound such as an alarm and they are used as warning signals.

5.2.5 Photodiodes can be used as a detector of infrared light in a control system.

(5 × 2) (10)

5.3 5.3.1 True
5.3.2 True
5.3.3 True
5.3.4 False
5.3.5 False
5.3.6 False
5.3.7 True

(7 × 1) (7)
**[22]**

**TOTAL: 100**

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